SYSDESIGN REFERENCE

Capacity Estimation Under Pressure

A senior-level reference with tradeoffs, failure modes, and interview follow-ups for capacity estimation under pressure.

Translate product usage into orders of magnitude for traffic, storage, bandwidth, and state.

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STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CAPACITY-ESTIMATION

SYSDESIGN

Capacity Estimation Under Pressure

CORE IDEA

Translate product usage into orders of magnitude for traffic, storage, bandwidth, and state.

Reference architecture decision chain

01	02	03	04
Peak Vs Average	Read/Write Mix	Payload Size	Retention

Decisions to defend

+ Peak vs average
+ read/write mix
+ payload size
+ retention
+ headroom
+ and growth.

Failure modes to test

+ False precision
+ missing replication
+ ignoring burstiness
+ and dimension errors.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Capacity Estimation Under Pressure

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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